

EXPERIMENT 5

Student Name: Jay Varshney

UID: 24BDA70206

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

Semester: 5

Subject Name: ADBMS

Subject Code: 24CSP-333

Experiment 5 -- To study and implement advanced SQL queries using Joins, Correlated Query's and Conditional statement.

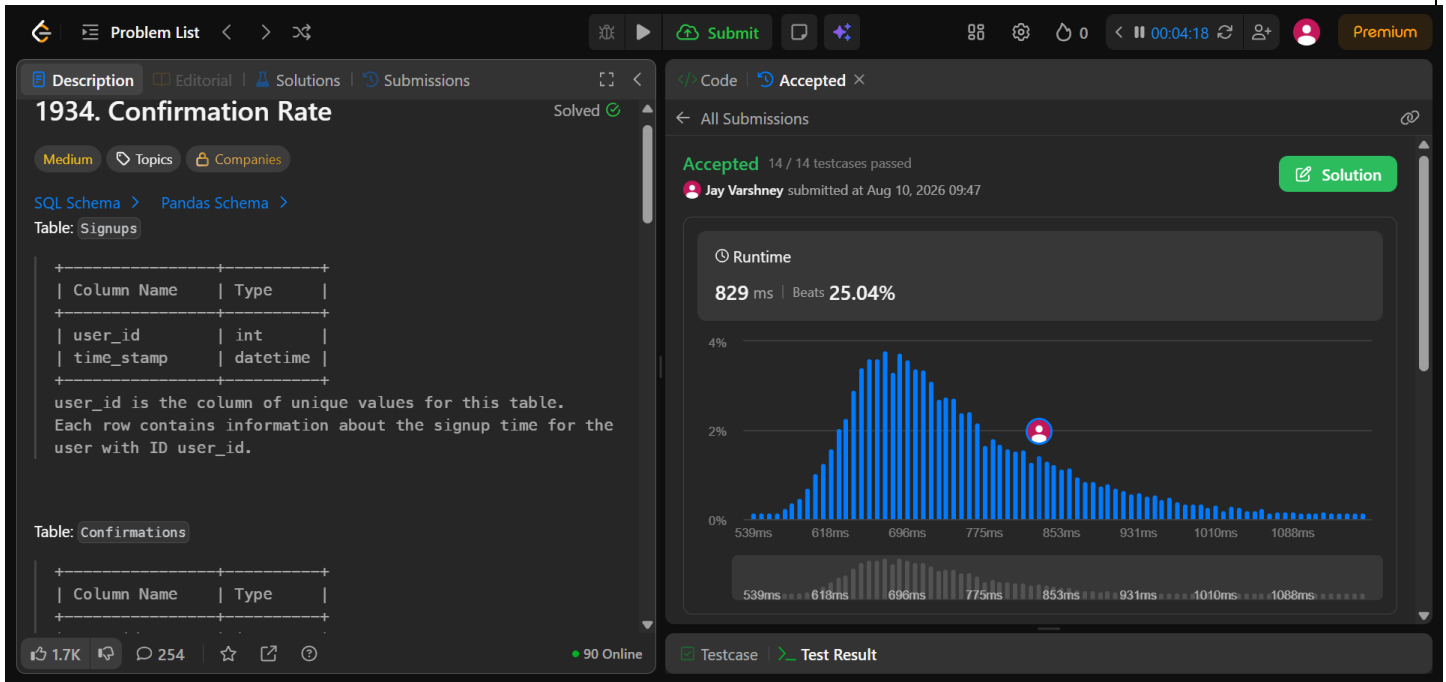
Aim : To study and implement correlated subqueries in SQL by retrieving employees whose salaries are among the top three highest distinct salary values within their respective departments, while correctly handling duplicate salary values using aggregate functions and joins

Question :5 Part A (i): Calculate Conformation Rate

SQL Query:

```
SELECT p.user_id, round(avg(CASE WHEN s.action like 'confirmed' THEN 1 ELSE 0 END),2) as  
confirmation_rate FROM confirmations as s right JOIN signups as p ON s.user_id = p.user_id group by p.user_id
```

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The screenshot shows a coding platform interface for a problem titled "1934. Confirmation Rate". The problem is marked as "Solved" and has a "Medium" difficulty level. The description includes a SQL schema for a table named "Signups" with columns "user_id" (int) and "time_stamp" (datetime). It also mentions a table named "Confirmations". The runtime for the submitted solution is 829 ms, which beats 25.04% of other solutions. A bar chart shows the distribution of runtime results for all submissions, with the user's solution marked by a red dot. The chart shows a peak around 696ms and a tail extending to 1088ms.

Question: Part A (ii) Calculate Cancellation rate SQL

Query:

```
SELECT tp.request_at as "Day", ROUND(AVG(CASE WHEN tp.status like 'cancelled%' THEN 1 ELSE 0
END),2)as "Cancellation Rate"
```

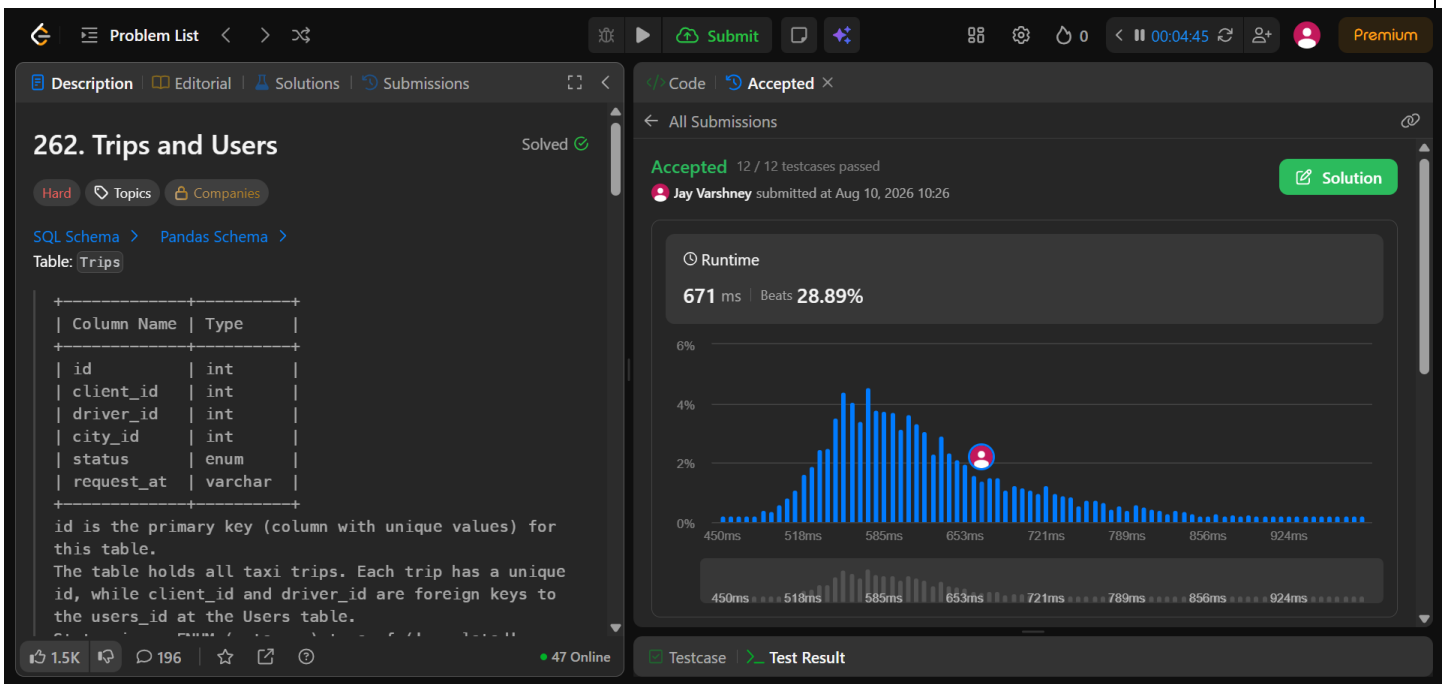
```
FROM trips as tp
```

```
JOIN Users as c ON c.users_id = tp.client_id
```

```
JOIN Users as d ON d.users_id = tp.driver_id
```

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WHERE c.banned = 'No' AND d.banned = 'No' AND tp.request_at BETWEEN '2013-10-01' AND '2013-10-03'
GROUP BY tp.request_at



The screenshot displays a coding platform interface for a problem titled "262. Trips and Users". The problem is marked as "Solved" and "Hard". The SQL Schema section shows the following table structure:

Column Name	Type
id	int
client_id	int
driver_id	int
city_id	int
status	enum
request_at	varchar

The description states: "id is the primary key (column with unique values) for this table. The table holds all taxi trips. Each trip has a unique id, while client_id and driver_id are foreign keys to the users_id at the Users table."

The right panel shows the submission status as "Accepted" with 12/12 testcases passed. The runtime performance is 671 ms, beating 28.89% of other submissions. A bar chart displays the distribution of runtime times across various testcases, with the user's submission highlighted at 671 ms.

Learning Outcome

After completing this experiment, the learner will be able to:

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1. **Understand and implement correlated subqueries** to solve SQL problems involving row-wise comparisons.
2. **Apply aggregate functions** such as COUNT() and DISTINCT to retrieve top-N records within groups.
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.
4. **Analyze and solve complex SQL queries** involving grouped data, ranking logic, and handling duplicate values effectively.